

Lecture 22

STAT 109: Introductory Biostatistics

Lecture 22

Normal Distribution and Standardization

Skills

- Compute a z-score given an observation, and the mean and standard deviation of a population.
- Interpret a z-score in context
- Use z-scores to perform standardized comparisons
- Compute the probability of a z-score or below using the Standard Normal Distribution and interpret it as the proportion of values below.

Example: Who is taller, Rosanna or Peter?

Two observations:

Rosanna's height in inches is _____

Peter's height in inches is _____

So _____ is taller when their heights are compared directly.

But who is taller relative to their group?

Draw a reference distribution for the stature of men and a reference distribution for the stature of women.

Identify the location of the center of the stature distribution on each plot and show one and two standard deviations (SD) on either side.

Add vertical lines to the appropriate distribution at Rosanna and Peter's heights. Who appears to be taller compared with the reference distributions?

According to the plots, _____ is taller relative to their group.

Instead of plotting two different reference distribution and comparing the observations where they fall relative to two SD from the center, we can compare the heights relative to their groups directly via **z-scores**.

Definition: z-score

$$z = \frac{\text{observation} - \text{mean of reference}}{\text{SD of reference}}$$

A z-score measures the number of SDs the observation is above the mean (or below if the difference is negative).

Compute the z-score for Rosanna's height using the reference information for women. Compute the z-score for Peter's

z-score using the reference information for men. Whose z-score is larger, that is, who is taller than more of their reference group?

If we assume the shape of a reference distribution is *Normal* then we can use a table or online calculator or R to find the percentage in the population that corresponds to a particular z-score.

Definition: Standard Normal

A standard normal distribution is a normal distribution with mean 0 and SD 1.

Notice that by the Empirical Rule, we can see that 68% of values are between -1 and 1, 95% are between -2 and 2, and 99% are between -3 and 3. The proportions of values corresponding to all those below a particular value are given in a **Standard Normal Table**:

[normal-table.pdf](#)

[Online Calculator](#)

Example: Rosanna's grip strength this morning using her dominant hand was 38.8 lbs. Suppose that mean grip strength is 66lbs and SD is 14lbs for women in their early forties in the US. Compute Rosanna's z-score and use it to find the percentage of women in their early forties in the US that Rosanna is stronger than assuming that this distribution is normal.

Example: Measure your own grip strength in lbs: _____

Compare with your choice of a reference group by computing your z-score.

Group	Mean (lbs)	SD (lbs)
College aged men	104	18
College aged women	62	16
Great Grandmothers	44	10

What proportion of your selected reference group are you stronger than? _____ Weaker than? _____

(use the online calculator or Standard Normal Table)